**PART A – THEORY**

Write this in a **Markdown** cell.

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**Data Analysis**

Data Analysis is the process of collecting, cleaning, transforming, and analyzing data to discover useful information, identify patterns, and support decision-making. It helps organizations understand their data and make informed business decisions.

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**Steps of a Data Science Project**

1. Problem Understanding
2. Data Collection
3. Data Cleaning
4. Exploratory Data Analysis (EDA)
5. Feature Engineering
6. Model Building
7. Model Evaluation
8. Deployment
9. Monitoring and Improvement

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**Machine Learning Problem Statement**

The objective of this project is to predict whether a customer is likely to purchase a product based on customer demographics, transaction history, and product information.

Target Variable:

**Purchased**

- 1 = Purchased
- 0 = Not Purchased

This is a **Binary Classification Problem**.

### What are Tensors?

A Tensor is a multidimensional array used to represent numerical data.

Examples:

- Scalar → 0D Tensor
- Vector → 1D Tensor
- Matrix → 2D Tensor
- 3D Array → 3D Tensor

TensorFlow and PyTorch use tensors for machine learning computations.

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### NumPy Tensor Examples

```
import numpy as np

scalar = np.array(10)

vector = np.array([1,2,3])

matrix = np.array([
   [1,2],
   [3,4]
])

tensor3d = np.array([
   [[1,2],[3,4]],
   [[5,6],[7,8]]
])

print("Scalar")
print(scalar)

print()

print("Vector")
print(vector)

print()
```

```
print("Matrix")
print(matrix)

print()

print("3D Tensor")
print(tensor3d)
```